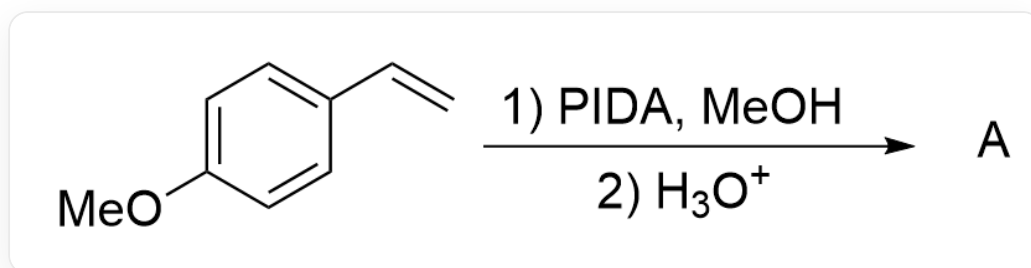
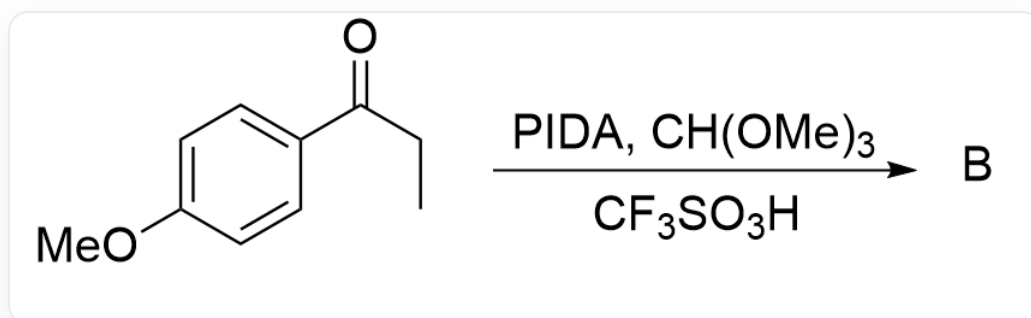


题目

PIDA是常见的相对温和的氧化剂。



C=CC1=CC=C(OC)C=C1>*PIDA.MeOH*> H_3O^+ >[A],A是反应产物

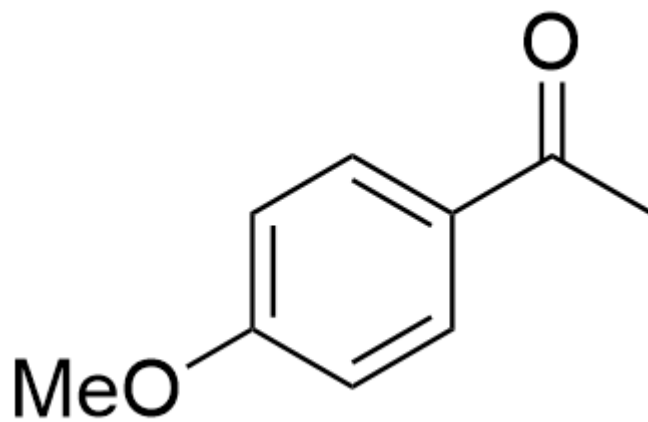


O=C(CC)C1=CC=C(OC)C=C1>*PIDA.CH(OMe)3.CF3SO3H*>[B],B是反应产物

请尝试预测以上两个反应的主产物

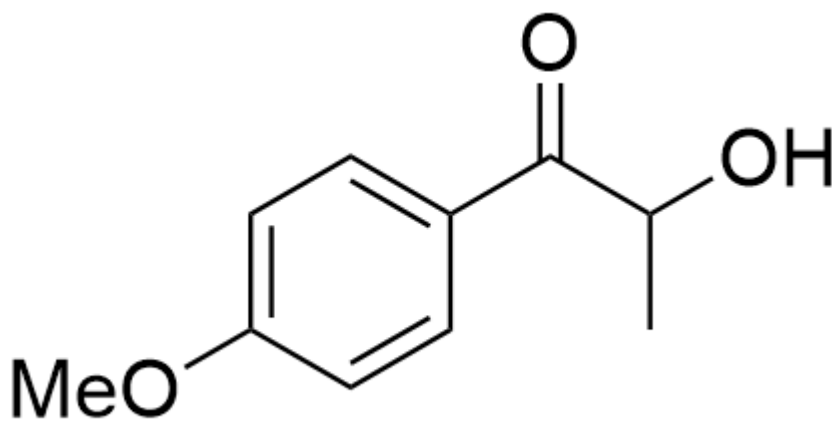
A. 其他选项均不正确

B.



CC(C1=CC=C(OC)C=C1)=O,产物A

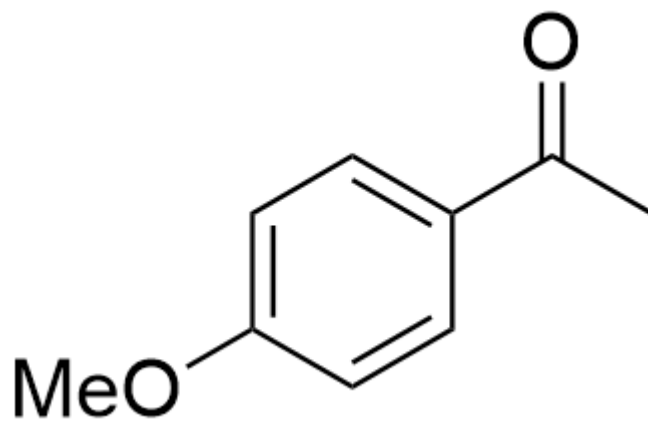
产物A



O=C(C(O)C)C1=CC=C(OC)C=C1,产物B

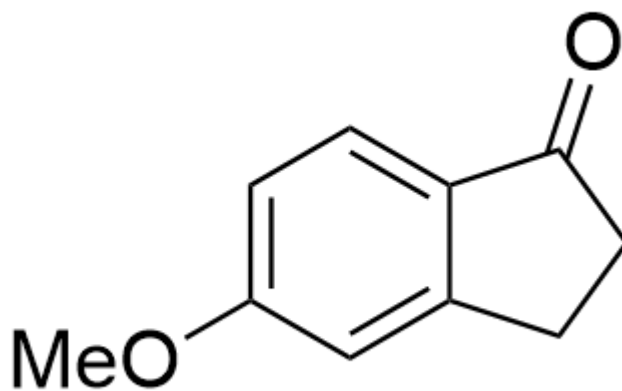
产物B

C.



CC(C1=CC=C(OC)C=C1)=O,产物A

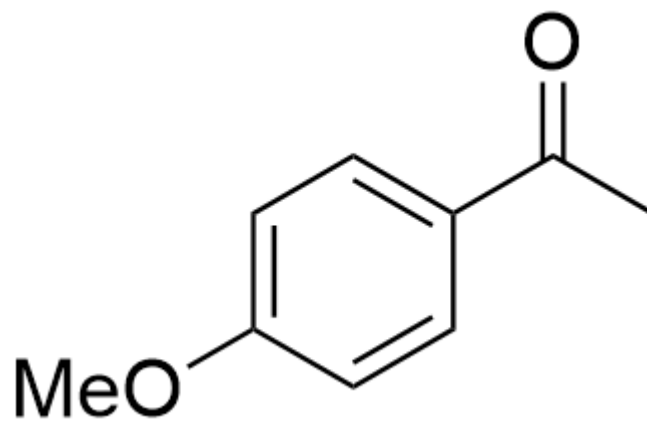
产物A



O=C1C2=CC=C(OC)C=C2CC1,产物B

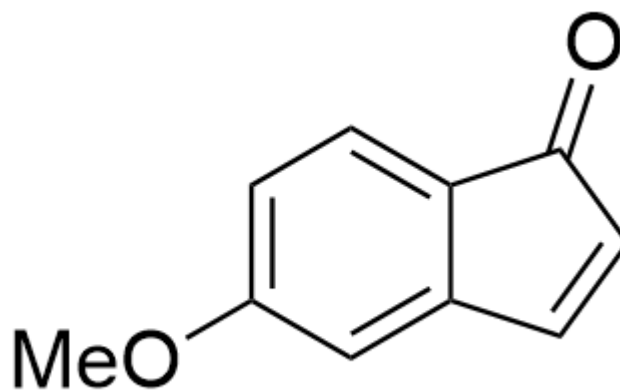
产物B

D.



CC(C1=CC=C(OC)C=C1)=O,产物A

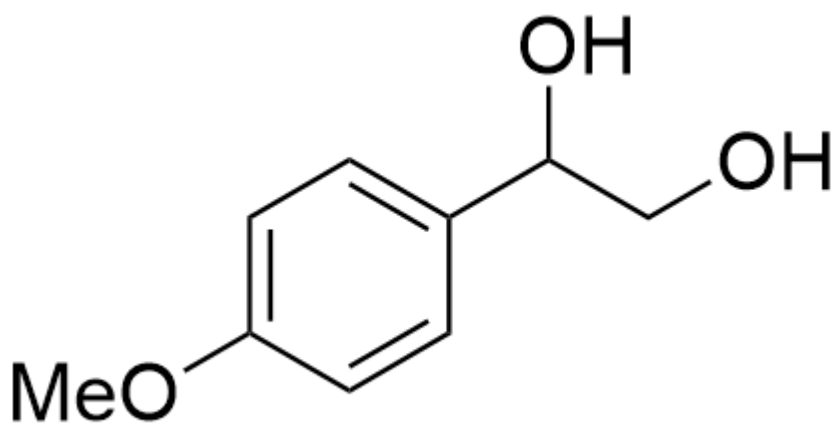
产物A



O=C1C2=CC=C(OC)C=C2C=C1,产物B

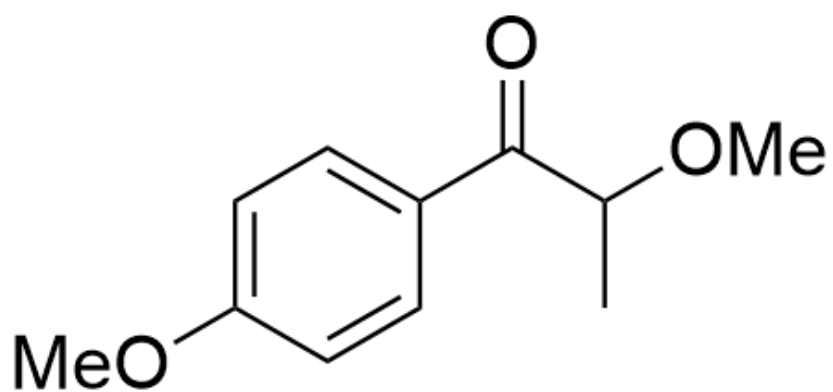
产物B

E.



OC(CO)C1=CC=C(OC)C=C1,产物A

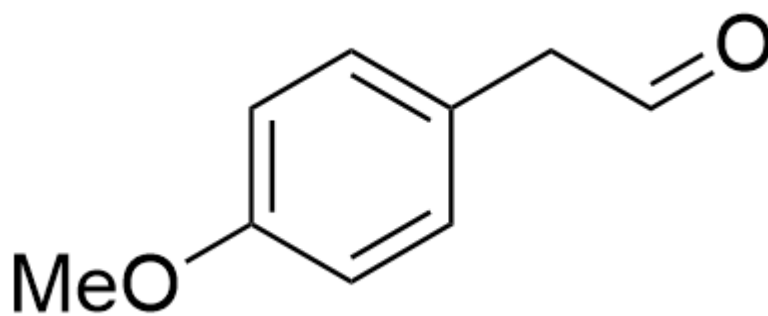
产物A



O=C(C(OC)C)C1=CC=C(OC)C=C1,产物B

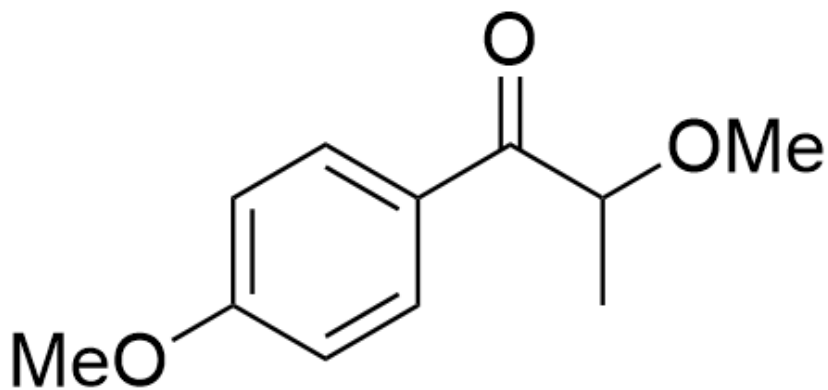
产物B

F.



O=CCC1=CC=C(OC)C=C1,产物A

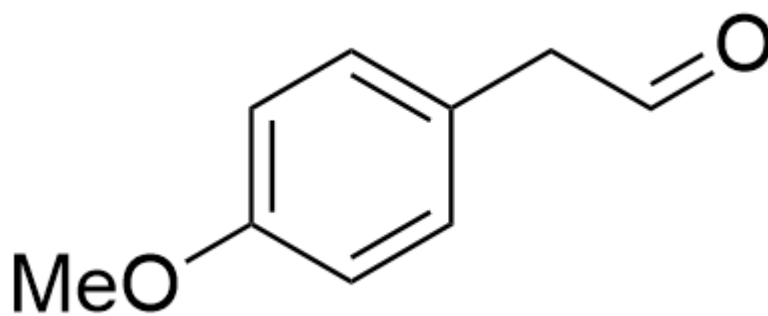
产物A



O=C(C(OC)C)C1=CC=C(OC)C=C1,产物B

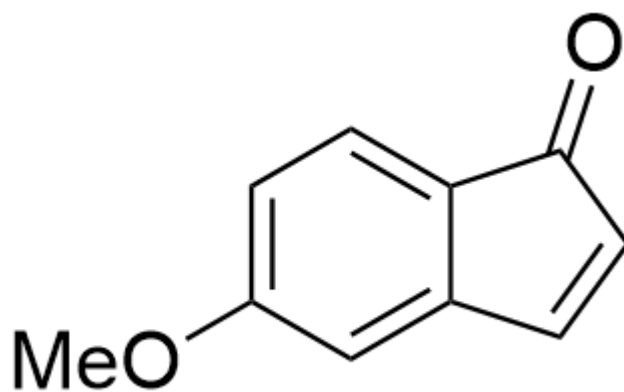
产物B

G.



O=CCC1=CC=C(OC)C=C1,产物A

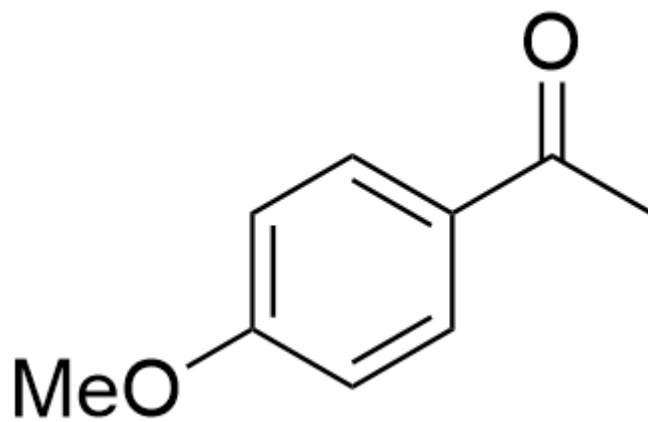
产物A



O=C1C2=CC=C(OC)C=C2C=C1,产物B

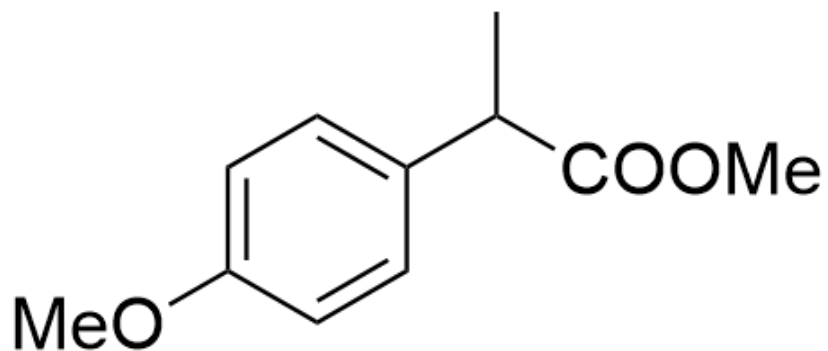
产物B

H.



CC(C1=CC=C(OC)C=C1)=O,产物A

产物A



CC(C(OC)=O)C1=CC=C(OC)C=C1,产物B

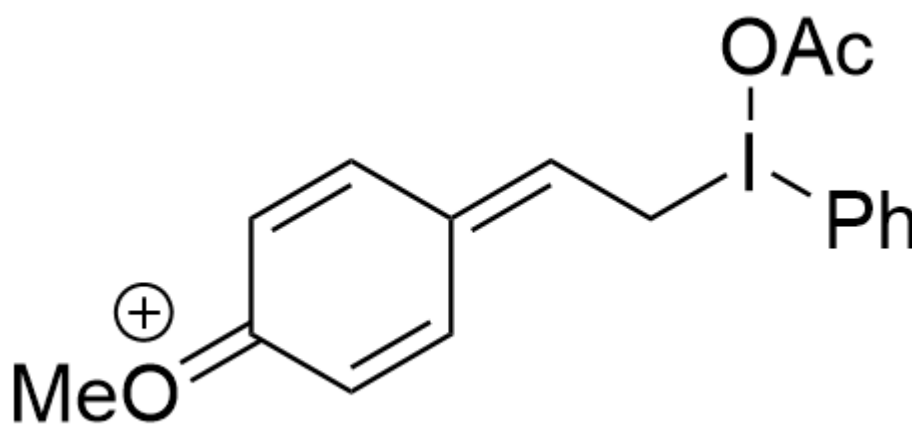
产物B

答案

正确答案: A

详细解析

对于反应1来说，底物首先与PIDA反应生成中间体



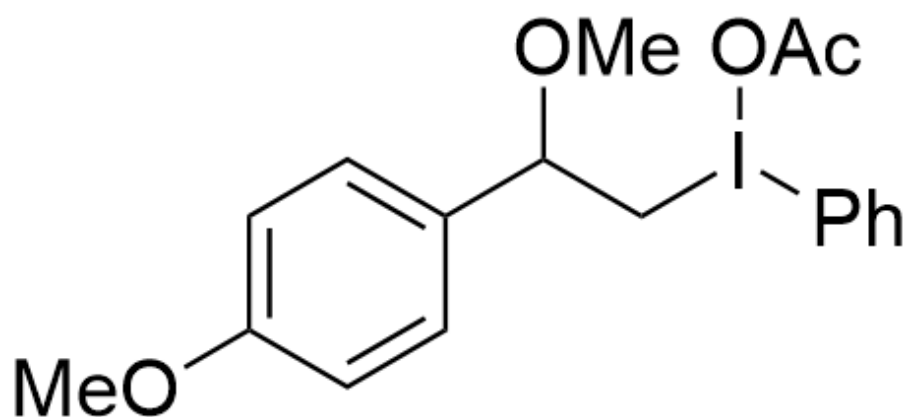
C[O+]=C(C=C/1)C=CC1=C\CI(C2=CC=CC=C2)OC(C)=O

CHECKPOINT

1 PTS

底物首先与PIDA反应生成中间体C[O+]=C(C=C/1)C=CC1=C\CI(C2=CC=CC=C2)OC(C)=O

接着可能与甲醇继续反应形成中间体



COC1=CC=C(C(OC)C(C2=CC=CC=C2)OC(C)=O)C=C1

CHECKPOINT

1 PTS

与甲醇继续反应形成中间体COC1=CC=C(C(OC)C(C2=CC=CC=C2)OC(C)=O)C=C1

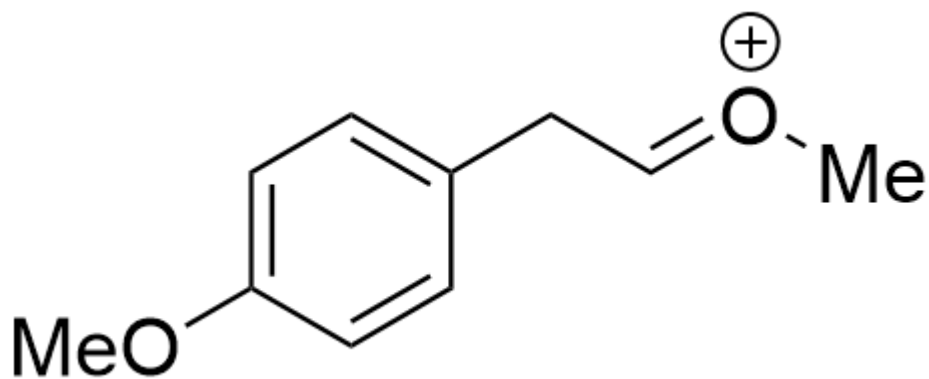
富电子的苯环优先于氢进行迁移

CHECKPOINT

1 PTS

富电子的苯环优先于氢进行迁移

得到中间体



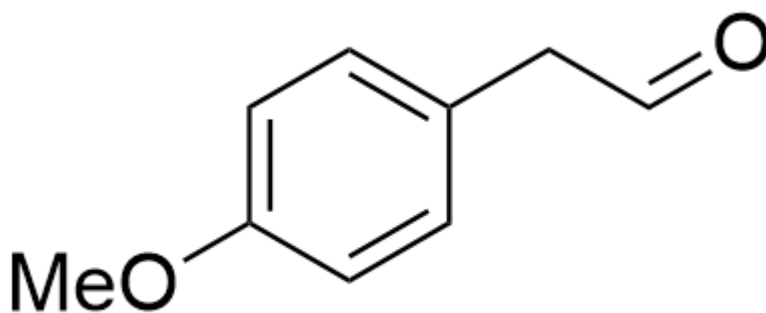
COC1=CC=C(C/C=[O+]/C)C=C1

CHECKPOINT

1 PTS

苯环迁移得到中间体COC1=CC=C(C/C=[O+]/C)C=C1

所得中间体进行水解，得到最终的醛类产物



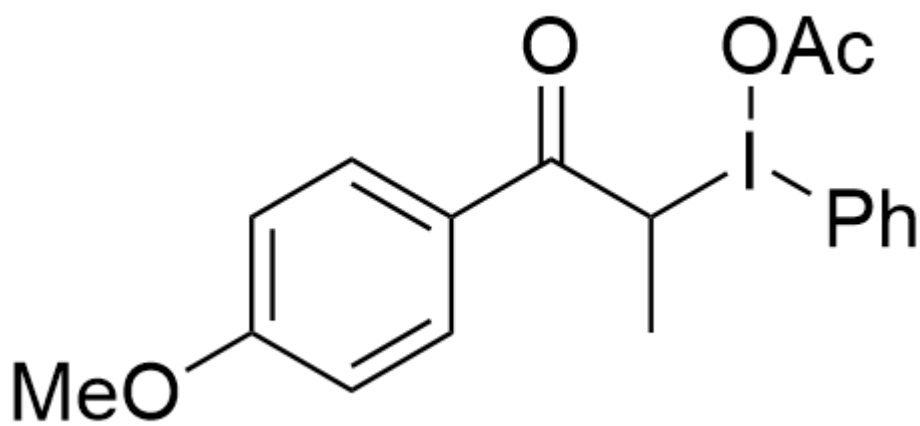
O=CCC1=CC=C(OC)C=C1

CHECKPOINT

2 PTS

中间体进行水解得到产物A, O=CCC1=CC=C(OC)C=C1

对于反应2, 首先可能得到中间体



O=C(C(I(OC(C)=O)C1=CC=CC=C1)C)C2=CC=C(OC)C=C2

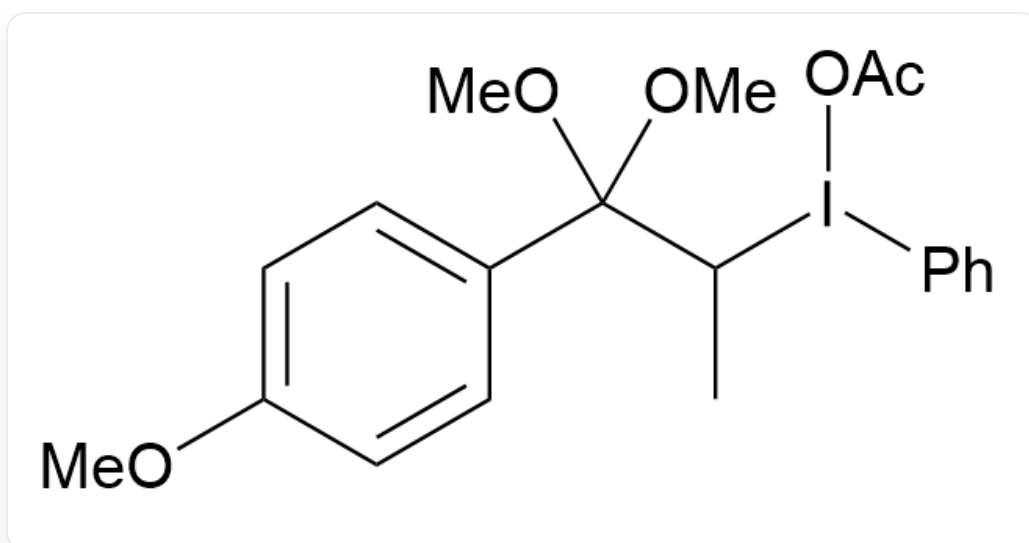
CHECKPOINT

1 PTS

中间体

O=C(C(I(OC(C)=O)C1=CC=CC=C1)C)C2=CC=C(OC)C=C2

随后进一步被酯化, 得到中间体



CC(I(OC(C)=O)C1=CC=CC=C1)C(OC)(OC)C2=CC=C(OC)C=C2

CHECKPOINT

1 PTS

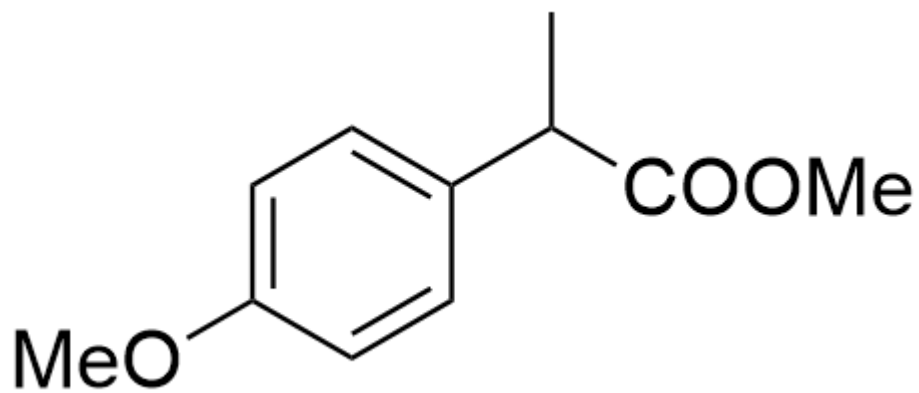
进一步酯化得到中间体CC(I(OC(C)=O)C1=CC=CC=C1)C(OC)(OC)C2=CC=C(OC)C=C2

最后发生苯环的迁移，经水解得到最终产物

CHECKPOINT

1 PTS

苯环迁移，水解得到最终产物**B**



CC(C(OC)=O)C1=CC=C(OC)C=C1,产物B

CHECKPOINT

2 PTS

产物B为CC(C(OC)=O)C1=CC=C(OC)C=C1

从而没有选项匹配结构，选项A正确。